1. Design a simple Android app using Kotlin basics. Display a welcome message on the screen. Use variables, data types, and control statements.

```
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    android:orientation="vertical">

    <EditText
        android:id="@+id/editTextName"
        android:layout_width="200dp"
        android:layout_height="wrap_content"
        android:hint="Enter your name"/>

    <Button
        android:id="@+id/button"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Submit"/>

    <TextView
        android:id="@+id/textView"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:textSize="22sp"
        android:text="Hello"/>
</LinearLayout>
```

```
package com.example.simpleapp

import android.os.Bundle
import android.widget.*
import androidx.appcompat.app.AppCompatActivity

class MainActivity : AppCompatActivity() {

    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_main)

        val textView = findViewById<TextView>(R.id.textView)
        val editText = findViewById<EditText>(R.id.editTextName)
        val button = findViewById<Button>(R.id.button)

        button.setOnClickListener {
            val name = editText.text.toString()

            if (name.isNotEmpty()) {
```

```
            textView.text = "Hello $name"
        } else {
            textView.text = "Please enter your name"
        }
    }
  }
}
```

2. Create an Android app to perform arithmetic operations. Use functions for addition, subtraction, multiplication, and division. Display results based on user input

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="20dp"
    android:gravity="center">

    <EditText
        android:id="@+id/num1"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter first number"
        android:inputType="numberDecimal"/>

    <EditText
        android:id="@+id/num2"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter second number"
        android:inputType="numberDecimal"/>

    <Button
        android:id="@+id/btnAdd"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Add"/>

    <Button
        android:id="@+id/btnSub"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Subtract"/>

    <Button
        android:id="@+id/btnMul"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
```

```
        android:text="Multiply"/>

    <Button
        android:id="@+id/btnDiv"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Divide"/>

    <TextView
        android:id="@+id/result"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Result"
        android:textSize="22sp"
        android:paddingTop="20dp"/>

</LinearLayout>
```

```
package com.example.simpleapp

import android.os.Bundle
import android.widget.*
import androidx.appcompat.app.AppCompatActivity

class MainActivity : AppCompatActivity() {

    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_main)

        val num1 = findViewById<EditText>(R.id.num1)
        val num2 = findViewById<EditText>(R.id.num2)

        val btnAdd = findViewById<Button>(R.id.btnAdd)
        val btnSub = findViewById<Button>(R.id.btnSub)
        val btnMul = findViewById<Button>(R.id.btnMul)
        val btnDiv = findViewById<Button>(R.id.btnDiv)

        val result = findViewById<TextView>(R.id.result)

        // ADD
        btnAdd.setOnClickListener {
            val a = num1.text.toString().toDouble()
            val b = num2.text.toString().toDouble()
            result.text = "Result: ${add(a, b)}"
        }

        // SUBTRACT
        btnSub.setOnClickListener {
```

```
            val a = num1.text.toString().toDouble()
            val b = num2.text.toString().toDouble()
            result.text = "Result: ${subtract(a, b)}"
        }

        // MULTIPLY
        btnMul.setOnClickListener {
            val a = num1.text.toString().toDouble()
            val b = num2.text.toString().toDouble()
            result.text = "Result: ${multiply(a, b)}"
        }

        // DIVIDE
        btnDiv.setOnClickListener {
            val a = num1.text.toString().toDouble()
            val b = num2.text.toString().toDouble()

            if (b != 0.0) {
                result.text = "Result: ${divide(a, b)}"
            } else {
                result.text = "Cannot divide by zero"
            }
        }
    }

    // FUNCTIONS

    fun add(a: Double, b: Double): Double {
        return a + b
    }

    fun subtract(a: Double, b: Double): Double {
        return a - b
    }

    fun multiply(a: Double, b: Double): Double {
        return a * b
    }

    fun divide(a: Double, b: Double): Double {
        return a / b
    }
}
```

3. Develop an app to manage student details. Use classes and objects in Kotlin. Display student information on the screen.

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="20dp"
    android:gravity="center">

    <EditText
        android:id="@+id/name"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter Name"/>

    <EditText
        android:id="@+id/age"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter Age"
        android:inputType="number"/>

    <EditText
        android:id="@+id/course"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter Course"/>

    <Button
        android:id="@+id/btnShow"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Show Details"/>

    <TextView
        android:id="@+id/result"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Student Info"
        android:textSize="20sp"
        android:paddingTop="20dp"/>

</LinearLayout>
```

```
class MainActivity : AppCompatActivity() {
class Student(
```

```
var name: String,
var age: Int,
var course: String
) {
fun getDetails(): String {
            return "Name: $name\nAge: $age\nCourse: $course"
      }
}
override fun onCreate(savedInstanceState: Bundle?) {
super.onCreate(savedInstanceState)
setContentView(R.layout.activity_main)

val name = findViewById<EditText>(R.id.name)
val age = findViewById<EditText>(R.id.age)
val course = findViewById<EditText>(R.id.course)
val button = findViewById<Button>(R.id.btnShow)
val result = findViewById<TextView>(R.id.result)

button.setOnClickListener {

val studentName = name.text.toString()
val studentAge = age.text.toString().toInt()
val studentCourse = course.text.toString()

// Creating Object
val student = Student(studentName, studentAge, studentCourse)

// Display Details
result.text = student.getDetails()
}
}
}
```

4. Build your first Android app. Add a button and handle click events. Show a toast message when clicked.

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
   android:layout_width="match_parent"
   android:layout_height="match_parent"
   android:gravity="center"
   android:orientation="vertical">

   <Button
      android:id="@+id/button"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="Click Me"/>
</LinearLayout>
package com.example.simpleapp
```

```
import android.os.Bundle
import android.widget.Button
import android.widget.Toast
import androidx.appcompat.app.AppCompatActivity

class MainActivity : AppCompatActivity() {

    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_main)

        val button = findViewById<Button>(R.id.button)

        button.setOnClickListener {
            Toast.makeText(this, "Button Clicked!", Toast.LENGTH_SHORT).show()
        }
    }
}
```

5. Design a login screen using layouts. Use LinearLayout or ConstraintLayout. Include username and password fields.

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="24dp"
    android:gravity="center">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Login"
        android:textSize="26sp"
        android:textStyle="bold"
        android:paddingBottom="20dp"/>

    <EditText
        android:id="@+id/username"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter Username"/>

    <EditText
        android:id="@+id/password"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
```

```
        android:hint="Enter Password"
        android:inputType="textPassword"
        android:layout_marginTop="10dp"/>

    <Button
        android:id="@+id/loginBtn"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Login"
        android:layout_marginTop="20dp"/>

</LinearLayout>
val username = findViewById<EditText>(R.id.username)
val password = findViewById<EditText>(R.id.password)
val button = findViewById<Button>(R.id.loginBtn)

button.setOnClickListener {
    Toast.makeText(this, "Login Clicked", Toast.LENGTH_SHORT).show()
}
```

6. Create navigation between two screens. Move from Login screen to Home screen. Use intents or navigation components.

activity_main.xml

```
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="24dp">

    <EditText
        android:id="@+id/username"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Username"/>

    <EditText
        android:id="@+id/password"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Password"
        android:inputType="textPassword"/>

    <Button
        android:id="@+id/loginBtn"
        android:layout_width="wrap_content"
```

```
        android:layout_height="wrap_content"
        android:text="Login"
        android:layout_marginTop="20dp"/>
</LinearLayout>
```

MainActivity.kt

```
package com.example.simpleapp

import android.content.Intent
import android.os.Bundle
import android.widget.*
import androidx.appcompat.app.AppCompatActivity

class MainActivity : AppCompatActivity() {

    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_main)

        val button = findViewById<Button>(R.id.loginBtn)

        button.setOnClickListener {

            // Intent to move to HomeActivity
            val intent = Intent(this, HomeActivity::class.java)
            startActivity(intent)
        }
    }
}
```

activity_home.xml

```
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Welcome to Home Screen"
        android:textSize="22sp"/>
</LinearLayout>
```

HomeActivity.kt

```
package com.example.simpleapp
```

```
import android.os.Bundle
import androidx.appcompat.app.AppCompatActivity

class HomeActivity : AppCompatActivity() {

    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_home)
    }
}
```

Add this in **AndroidManifest.xml**

```
None
<activity android:name=".HomeActivity" />
```

7. Demonstrate Activity and Fragment lifecycle. Log each lifecycle method. Display lifecycle changes on screen.

activity_main.xml

```
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="16dp">

    <TextView
        android:id="@+id/activityText"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Activity Lifecycle"
        android:textSize="18sp"
        android:padding="10dp"/>

    <FrameLayout
        android:id="@+id/fragmentContainer"
        android:layout_width="match_parent"
        android:layout_height="200dp"/>
</LinearLayout>
```

MainActivity.kt

```
package com.example.simpleapp
```

```kotlin
import android.os.Bundle
import android.util.Log
import android.widget.TextView
import androidx.appcompat.app.AppCompatActivity

class MainActivity : AppCompatActivity() {

    lateinit var textView: TextView

    private fun update(msg: String) {
        Log.d("ACTIVITY", msg)
        textView.append("\n$msg")
    }

    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_main)

        textView = findViewById(R.id.activityText)
        update("onCreate")

        // Load Fragment
        supportFragmentManager.beginTransaction()
            .replace(R.id.fragmentContainer, MyFragment())
            .commit()
    }

    override fun onStart() {
        super.onStart()
        update("onStart")
    }

    override fun onResume() {
        super.onResume()
        update("onResume")
    }

    override fun onPause() {
        super.onPause()
        update("onPause")
    }

    override fun onStop() {
        super.onStop()
        update("onStop")
    }

    override fun onDestroy() {
        super.onDestroy()
```

```
        update("onDestroy")
    }
}
```

fragment_my.xml

```
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical">

    <TextView
        android:id="@+id/fragmentText"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Fragment Lifecycle"
        android:padding="10dp"/>
</LinearLayout>
```

MyFragment.kt

```
package com.example.simpleapp

import android.os.Bundle
import android.util.Log
import android.view.*
import android.widget.TextView
import androidx.fragment.app.Fragment

class MyFragment : Fragment() {

    lateinit var textView: TextView

    private fun update(msg: String) {
        Log.d("FRAGMENT", msg)
        textView.append("\n$msg")
    }

    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        Log.d("FRAGMENT", "onCreate")
    }

    override fun onCreateView(
        inflater: LayoutInflater, container: ViewGroup?,
        savedInstanceState: Bundle?
    ): View {
        val view = inflater.inflate(R.layout.fragment_my, container, false)
```

```
        textView = view.findViewById(R.id.fragmentText)
        update("onCreateView")
        return view
    }

    override fun onStart() {
        super.onStart()
        update("onStart")
    }

    override fun onResume() {
        super.onResume()
        update("onResume")
    }

    override fun onPause() {
        super.onPause()
        update("onPause")
    }

    override fun onStop() {
        super.onStop()
        update("onStop")
    }

    override fun onDestroy() {
        super.onDestroy()
        update("onDestroy")
    }
}
```

8. Develop an app using MVVM architecture. Create UI layer with ViewModel. Display a list of items.

```
com.example.simpleapp
│
├── model/
│   Item.kt
│
├── viewmodel/
│   ItemViewModel.kt
│
└── MainActivity.kt
```

Item.kt

```
package com.example.simpleapp.model

data class Item(
    val name: String
)
```

ItemViewModel.kt

```
package com.example.simpleapp.viewmodel

import androidx.lifecycle.ViewModel
import com.example.simpleapp.model.Item

class ItemViewModel : ViewModel() {

    fun getItems(): List<Item> {
        return listOf(
            Item("Apple"),
            Item("Banana"),
            Item("Orange"),
            Item("Mango"),
            Item("Grapes")
        )
    }
}
```

activity_main.xml

```
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <ListView
        android:id="@+id/listView"
        android:layout_width="match_parent"
        android:layout_height="match_parent"/>
</LinearLayout>
```

MainActivity (View)

```
package com.example.simpleapp

import android.os.Bundle
import android.widget.ArrayAdapter
import android.widget.ListView
import androidx.appcompat.app.AppCompatActivity
import androidx.lifecycle.ViewModelProvider
import com.example.simpleapp.viewmodel.ItemViewModel

class MainActivity : AppCompatActivity() {

    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_main)
```

```
        val listView = findViewById<ListView>(R.id.listView)

        // Get ViewModel
        val viewModel = ViewModelProvider(this).get(ItemViewModel::class.java)

        // Get data
        val items = viewModel.getItems()

        // Convert to simple list of strings
        val itemNames = items.map { it.name }

        // Set adapter
        val adapter = ArrayAdapter(this, android.R.layout.simple_list_item_1, itemNames)
        listView.adapter = adapter
    }
}
```

```
View (Activity)
  ↓
ViewModel provides data
  ↓
Model holds data
  ↓
UI displays list
```

9. Build an app with data persistence. Use Room database to store data. Retrieve and display stored data.

```
model/
   Student.kt
dao/
   StudentDao.kt
database/
   AppDatabase.kt
MainActivity.kt
```

Student.kt

```
package com.example.simpleapp.model

import androidx.room.Entity
import androidx.room.PrimaryKey

@Entity
data class Student(
   @PrimaryKey(autoGenerate = true)
   val id: Int = 0,
   val name: String
)
```

StudentDao.kt

```kotlin
package com.example.simpleapp.dao

import androidx.room.Dao
import androidx.room.Insert
import androidx.room.Query
import com.example.simpleapp.model.Student

@Dao
interface StudentDao {

    @Insert
    fun insert(student: Student)

    @Query("SELECT * FROM Student")
    fun getAll(): List<Student>
}
```

AppDatabase.kt

```kotlin
package com.example.simpleapp.database

import androidx.room.Database
import androidx.room.RoomDatabase
import com.example.simpleapp.dao.StudentDao
import com.example.simpleapp.model.Student

@Database(entities = [Student::class], version = 1)
abstract class AppDatabase : RoomDatabase() {
    abstract fun studentDao(): StudentDao
}
```

XML Layout

```xml
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="16dp">

    <EditText
        android:id="@+id/nameInput"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter Name"/>

    <Button
```

```
        android:id="@+id/addBtn"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Add"/>

    <Button
        android:id="@+id/showBtn"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Show All"/>

    <TextView
        android:id="@+id/result"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:textSize="18sp"
        android:paddingTop="20dp"/>
</LinearLayout>
```

MainActivity

```
package com.example.simpleapp

import android.os.Bundle
import android.widget.*
import androidx.appcompat.app.AppCompatActivity
import androidx.room.Room
import com.example.simpleapp.database.AppDatabase
import com.example.simpleapp.model.Student

class MainActivity : AppCompatActivity() {

    lateinit var db: AppDatabase

    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_main)

        val input = findViewById<EditText>(R.id.nameInput)
        val addBtn = findViewById<Button>(R.id.addBtn)
        val showBtn = findViewById<Button>(R.id.showBtn)
        val result = findViewById<TextView>(R.id.result)

        // Create DB
        db = Room.databaseBuilder(
            applicationContext,
            AppDatabase::class.java,
            "student_db"
```

```
        ).allowMainThreadQueries() // for simplicity
         .build()

        val dao = db.studentDao()

        // Insert
        addBtn.setOnClickListener {
            val name = input.text.toString()
            if (name.isNotEmpty()) {
                dao.insert(Student(name = name))
                Toast.makeText(this, "Saved!", Toast.LENGTH_SHORT).show()
                input.text.clear()
            }
        }

        // Retrieve
        showBtn.setOnClickListener {
            val students = dao.getAll()
            val text = students.joinToString("\n") { it.name }
            result.text = text
        }
    }
}
```

10. Create an app using RecyclerView. Display a list of items with images. Handle item click events.

```
com.example.simpleapp
│
├── Item.kt
├── ItemAdapter.kt
└── MainActivity.kt

res/layout
├── activity_main.xml
└── item_row.xml
```

Item.kt

```
package com.example.simpleapp

data class Item(
    val name: String,
    val image: Int
)
```

item_row.xml

```
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:orientation="horizontal"
    android:padding="10dp"
```

```xml
    android:layout_width="match_parent"
    android:layout_height="wrap_content">

    <ImageView
        android:id="@+id/image"
        android:layout_width="60dp"
        android:layout_height="60dp"/>

    <TextView
        android:id="@+id/name"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:textSize="18sp"
        android:layout_marginStart="10dp"/>
</LinearLayout>
```

ItemAdapter.kt

```kotlin
package com.example.simpleapp

import android.view.*

import android.widget.*

import androidx.recyclerview.widget.RecyclerView

class ItemAdapter(

    private val list: List<Item>

) : RecyclerView.Adapter<ItemAdapter.ViewHolder>() {

    class ViewHolder(view: View) : RecyclerView.ViewHolder(view) {

        val image: ImageView = view.findViewById(R.id.image)

        val name: TextView = view.findViewById(R.id.name)

    }

    override fun onCreateViewHolder(parent: ViewGroup, viewType: Int): ViewHolder {

        val view = LayoutInflater.from(parent.context)

            .inflate(R.layout.item_row, parent, false)

        return ViewHolder(view)

    }
```

```
    override fun onBindViewHolder(holder: ViewHolder, position: Int) {

        val item = list[position]

        holder.name.text = item.name

        holder.image.setImageResource(item.image)

        // Click handling

        holder.itemView.setOnClickListener {

            Toast.makeText(holder.itemView.context, item.name, Toast.LENGTH_SHORT).show()

        }

    }

    override fun getItemCount(): Int = list.size

}
```

activity_main.xml

```
<androidx.recyclerview.widget.RecyclerView xmlns:android="http://schemas.android.com/apk/res/android"
    android:id="@+id/recyclerView"
    android:layout_width="match_parent"
    android:layout_height="match_parent"/>
```

MainActivity.kt

```
package com.example.simpleapp

import android.os.Bundle
import androidx.appcompat.app.AppCompatActivity
import androidx.recyclerview.widget.LinearLayoutManager
import androidx.recyclerview.widget.RecyclerView

class MainActivity : AppCompatActivity() {

    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_main)

        val recyclerView = findViewById<RecyclerView>(R.id.recyclerView)

        val list = listOf(
            Item("Apple", android.R.drawable.ic_menu_gallery),
            Item("Banana", android.R.drawable.ic_menu_camera),
            Item("Mango", android.R.drawable.ic_menu_compass),
            Item("Orange", android.R.drawable.ic_menu_call)
```

```
        )

        recyclerView.layoutManager = LinearLayoutManager(this)
        recyclerView.adapter = ItemAdapter(list)
    }
}
```

Data list
  ↓
Adapter
  ↓
RecyclerView
  ↓
Display items
  ↓
Click → Toast

11. Develop an app to connect to the internet. Fetch data from an API. Display the fetched data.

In `AndroidManifest.xml`:

```None
<uses-permission android:name="android.permission.INTERNET"/>
```

XML Layout

```
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    android:orientation="vertical"
    android:padding="16dp">

    <Button
        android:id="@+id/fetchBtn"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Fetch Data"/>

    <TextView
        android:id="@+id/result"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Result"
        android:paddingTop="20dp"/>
</LinearLayout>
```

MainActivity.kt

```
package com.example.simpleapp

import android.os.Bundle
import android.widget.*
import androidx.appcompat.app.AppCompatActivity
import org.json.JSONObject
import java.net.URL
import kotlin.concurrent.thread

class MainActivity : AppCompatActivity() {

    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_main)

        val button = findViewById<Button>(R.id.fetchBtn)
        val result = findViewById<TextView>(R.id.result)

        button.setOnClickListener {

            thread {
                try {
                    val response = URL("https://jsonplaceholder.typicode.com/todos/1").readText()
                    val json = JSONObject(response)

                    val title = json.getString("title")

                    runOnUiThread {
                        result.text = "Title: $title"
                    }

                } catch (e: Exception) {
                    runOnUiThread {
                        result.text = "Error fetching data"
                    }
                }
            }
        }
    }
}
```

12. Implement Repository Pattern. Use WorkManager for background tasks. Fetch and cache data efficiently.

```
com.example.simpleapp
│
├── DataRepository.kt
├── MyWorker.kt
```

├── MainActivity.kt

Dependencies

```None
implementation "androidx.work:work-runtime-ktx:2.9.0"
```

DataRepository.kt

```
package com.example.simpleapp

import java.net.URL

object DataRepository {

    var cachedData: String = ""

    fun fetchData(): String {
        val data = URL("https://jsonplaceholder.typicode.com/todos/1").readText()
        cachedData = data   // cache
        return data
    }

    fun getCached(): String {
        return cachedData
    }
}
```

MyWorker.kt

```
package com.example.simpleapp

import android.content.Context
import androidx.work.Worker
import androidx.work.WorkerParameters

class MyWorker(context: Context, params: WorkerParameters) :
    Worker(context, params) {

    override fun doWork(): Result {
        return try {
            DataRepository.fetchData()
            Result.success()
        } catch (e: Exception) {
            Result.failure()
        }
    }}
```

MainActivity

```kotlin
package com.example.simpleapp

import android.os.Bundle
import android.widget.*
import androidx.appcompat.app.AppCompatActivity
import androidx.work.OneTimeWorkRequestBuilder
import androidx.work.WorkManager

class MainActivity : AppCompatActivity() {

    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_main)

        val button = findViewById<Button>(R.id.button)
        val text = findViewById<TextView>(R.id.result)

        button.setOnClickListener {

            // Run background task
            val work = OneTimeWorkRequestBuilder<MyWorker>().build()
            WorkManager.getInstance(this).enqueue(work)

            // Show cached data
            text.text = DataRepository.getCached()
        }
    }
}
```

XML

```xml
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:gravity="center"
    android:orientation="vertical"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <Button
        android:id="@+id/button"
        android:text="Fetch Data"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"/>

    <TextView
        android:id="@+id/result"
        android:text="Result"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"/>
</LinearLayout>
```

What is Repository?

Middle layer between UI and data

What is WorkManager?

Runs background tasks reliably

What is caching?

Storing data locally to reuse

13. Design a user-friendly app UI. Apply colors, spacing, and alignment. Ensure good usability and user experience.

```
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="24dp"
    android:gravity="center"
    android:background="#F5F5F5">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Welcome"
        android:textSize="28sp"
        android:textStyle="bold"
        android:textColor="#333333"
        android:layout_marginBottom="20dp"/>

    <EditText
        android:id="@+id/nameInput"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter your name"
        android:padding="12dp"
        android:backgroundTint="#6200EE"/>

    <Button
        android:id="@+id/button"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Submit"
        android:layout_marginTop="20dp"
        android:backgroundTint="#6200EE"
        android:textColor="#FFFFFF"/>
```

```
    <TextView
        android:id="@+id/result"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Result"
        android:textSize="18sp"
        android:layout_marginTop="20dp"
        android:textColor="#555555"/>
</LinearLayout>
```

Simple Functionality (Optional)

```
val input = findViewById<EditText>(R.id.nameInput)

val button = findViewById<Button>(R.id.button)

val result = findViewById<TextView>(R.id.result)

button.setOnClickListener {

val name = input.text.toString()

result.text = "Hello $name"

}
```